Secur-Fi: A Secure Wireless Sensing System Based on Commercial Wi-Fi Devices

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***Abstract*—Wi-Fi sensing technology plays an important role in numerous IoT applications such as virtual reality, smart homes and elder healthcare. The basic principle is to extract physical features from the Wi-Fi signals to depict the user's locations or behaviors. However, current research focuses more on improving the sensing accuracy but neglects the security concerns. Specifically, current Wi-Fi router usually transmits a strong signal, so that we can access the Internet even through the wall. Accordingly, the outdoor adversaries are able to eavesdrop on this strong Wi-Fi signal, and infer the behavior of indoor users in a non-intrusive way, while the indoor users are unaware of this eavesdropping. To prevent outside eavesdropping, we propose Secur-Fi, a secure Wi-Fi sensing system. Our system meets the following two requirements: (1) we can generate fraud signals to block outside unauthorized Wi-Fi sensing; (2) we can recover the signal, and enable authorized Wi-Fi sensing. We implement the proposed system on commercial Wi-Fi devices and conduct experiments in three applications including passive tracking, behavior recognition, and breath detection. The experiment results show that our proposed approaches can reduce the accuracy of unauthorized sensing by 130% (passive tracking), 72% (behavior recognition), 86% (breath detection).**

***Index Terms*—Wi-Fi, Wireless Sensing, Channel State Information**

I. Introduction

A. Background and Motivation

The Wi-Fi sensing technology can remotely monitor the status of the user without devices [1]–[3], which brings many smart applications such as breath detection [4], [5] and activity recognition [6]–[8] to smart home and smart medical scenarios [9]. In practice, since there are usually multiple rooms in a home, the obstacles including walls and furniture block the propagation of Wi-Fi signals, and therefore reduce the accuracy of wireless sensing. To address this problem, existing researches explore a series of methods [10], [11] to extract fine-grained physical features, even when the wireless signal has been blocked by the wall. Hence, through-wall wireless sensing proves to be feasible.

Regardless of the high accuracy and convenience, these through-wall sensing technologies also cause serious security risks. Fig. 1 illustrates the basic idea of Wi-Fi sensing and the security risks. Specifically, our behavior influences the Wi-Fi signals reflected by our body and therefore changes the feature of the actual received signals. The basic idea of Wi-Fi sensing is to parse and infer user behavior from such

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Fig. 1: The attacker outside can pick up the Wi-Fi signal from the room to determine what is going on inside.

signal features. Currently, the Wi-Fi router usually transmits a stronger signal, so that we can access the Internet even through the wall. Accordingly, the outdoor adversaries are able to eavesdrop on Wi-Fi signals. Next, based on the through-wall sensing algorithm, the adversaries can infer the behavior of indoor users in a non-intrusive way, while the indoor users are unaware of this eavesdropping.

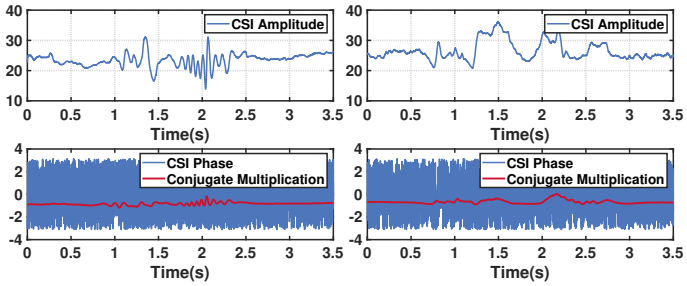
However, the existing researches pay less attention to security concerns. First, many wireless sensing methods are open access such as [12], [13]. That is, wireless sensing technology is available to both developers and adversaries. Second, Openwifi [14] imposes the artificial channel response of Wi-Fi signal to realize anti-sensing. Based on the proposed tool, only the authorized device can eliminate the artificial signal to obtain the actual channel response. However, Openwifi requires expensive hardware to realize the proposed function, which limits the deployment of the proposed system. To our knowledge, there is no other way to resist eavesdropping by attackers. As through-wall sensing proves to work, it is necessary to protect indoor users from this potential threat.

B. Challenges and Contributions

In this paper, we develop a secure Wi-Fi device to realize anti-monitoring. With the proposed system, users can still benefit from wireless sensing applications, while attackers cannot infer the indoor situation through walls. The basic idea is to change the features of the Wi-Fi signal by switching the transmitting antenna so that the outdoor eavesdropper cannot extract user's behaviors. Meanwhile, we record the rules of antenna switching and recover the signal for authorized indoor users. When implementing the proposed system, we will face the following challenges.

First, since current work [15] proves that there is a mapping relationship between the user's behavior and the signal

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(a) Sit (b) Squat

Fig. 2: The pattern of CSI amplitude and phase readings.

pattern, how to design the system so that the attacker can observe the signal that does not match the behavior? To address this challenge, our inspirations come from SWAN [16]. SWAN proposes a prototype based on antenna switching to control Wi-Fi signal transmission. When different antennas are enabled, the received signal strength is different. Hence, SWAN is able to find the signal with the best quality for communication and localization. Accordingly, if we design the rules of antenna switching reasonably, we are able to generate a varying signal even for static scenarios. Such a switching rule is called *behavior codebooks* in our system. Since the behavior codebook is the predefined rule at the transmitter, the adversary would not acquire it.

Second, if we change the signal pattern to prevent outside eavesdropping, how can we ensure the wireless sensing of the authorized user simultaneously? To address this challenge, the direct solution is to utilize the behavior codebook to recover the disturbing signal. However, our experiments find that there is usually an unpredictable delay in the change of the signal features after we perform the switching. To this end, we design a novel algorithm to take time delay, behavior codebook and features into account. In particular, we design two sliding windows to handle time delay, and propose circular standard deviation to determine the mutation point of the feature. Based on the proposed method, we can answer whether the CSI change is caused by the user's real behavior or the antenna switching. This solution makes it possible for the indoor receiver to recover the signal, thereby realizing authorized wireless sensing.

The contribution of this paper is as follows:

- We propose the Secur-Fi system. The proposed system is capable of preventing attackers from unauthorized wireless sensing, while maintaining authorized wireless sensing. To the best of our knowledge, Secur-Fi is the first system to prevent unauthorized wireless sensing based on the commercial Wi-Fi devices.
- In order to ensure authorized wireless sensing, we combine the sliding window, behavior codebook and circular standard deviation to design the signal recovery method. The proposed method can solve both the delay problem of antenna switching and periodic problem of CSI phase.
- We implement Secur-Fi and conduct experiments to evaluate the performance of Secur-Fi. Experimental results show that the proposed system is able to prevent attackers from passive tracking, breath detection, and behavior recognition.

The remainder of this paper is organized as follows. We introduce the preliminary knowledge and threat model in §Section. II. §Section. III gives a basic model of the proposed system, and §Section. IV presents our approach in detail. Next, we implement the Secur-Fi and conduct comprehensive experiments to verify the proposed system in §Section. V. We discuss the related work in §Section. VI, and finally conclude the paper in §Section. VII.

II. PRELIMINARY

A. CSI Premier

Channel state information (CSI) reflects the changes in the amplitude and phase of the wireless signal during the signal propagation. In practice, the received signal is a superposition of signals from multiple paths, which is known as multi-path propagation. Mathematically, CSI with the frequency of f at time t can be expressed as:

$$H(f,t) = \sum_{n=1}^{N} \alpha_n(t) e^{-j2\pi \frac{d_n(t)}{\lambda}}, \tag{1}$$

where N, $\alpha_n(t)$ and $d_n(t)$ are the path number, complex attenuation factor and the propagation length of the n-th path.

According to the prior work [6], the paths can be grouped into static paths and dynamic paths. The dynamic component is the signal reflected by moving objects such as the human body, while the static component consists of line-of-sight (LoS) propagation and the signal reflected by the static objects in the environment. If we utilize H_s and G_m to respectively represent the static and dynamic components, CSI can be rewritten as

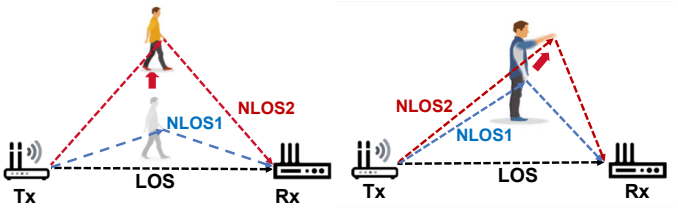
$$\begin{aligned} H(f,t) &= H_s(f,t) + H_d(f,t) \\ &= H_s(f,t) + \sum_{n \in G_m} \alpha_n(f,t) e^{-j2\pi \frac{d_n(t)}{\lambda}}, \end{aligned} \tag{2}$$

where $2\pi \frac{d(t)}{\lambda}$ is the phase shift caused by the moving objects. It is obvious that the user's behavior will affect the dynamic path signal reflected by the human. This effect is reflected in changes in the amplitude and phase of the received signal. In practice, we can only observe the distinct pattern in amplitude rather than phase, as shown in Fig. 2. It is because phase readings are affected by carrier frequency offset (CFO) and sample frequency offset (SFO). Specifically, the actual CSI is composed of signals affected by the environment and phase noise caused by imperfect hardware designs. In this case, CSI can be expressed as

$$H(f,t) = e^{-j\phi_e}(H_s(f,t) + \sum_{n \in G_m} \alpha_n(f,t) e^{-j2\pi \frac{d_n(t)}{\lambda}}), \tag{3}$$

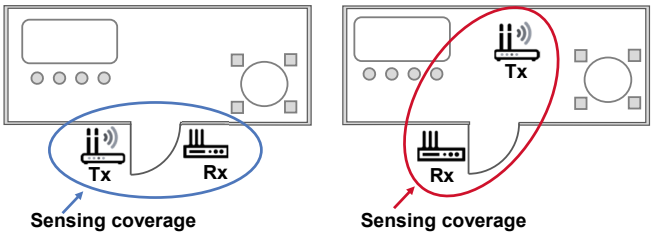
where ϕ_e is the time-variant random phase offset. According to Eq. 3, the phase noise disguises the actual phase information. To solve this problem, the arts [17] utilize phase difference as features to depict user's behavior. There are two common methods to resolve phase differences, *i.e.*, conjugate multiplication [12] and CSI radio [4], [6].

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(a) Walk (b) Change gestures

Fig. 3: User actions change the length of reflection paths.



(a) (b)

Fig. 4: The range of the sensing area is related to the placement of the transceivers.

The conjugate multiplication is a well-known method to remove phase offset. The basic idea is that phase noise generated by CFO is different in the frequency domain and time domain, and there is no difference across different antennas [18]. Therefore, we select a reference antenna, and the conjugate of the CSI with respect to this antenna is denoted as

$$\overline{H}_0(f,t) = e^{j\theta_{offset}}(H_{s_0} + \sum_{n_0 \in G_{m_0}} \alpha_{n_0}(f,t)e^{-j2\pi\frac{d_{n_0}(t)}{\lambda}}). \tag{4}$$

We can calculate the conjugate multiplication between the reference antenna and each antenna to remove random CSI phase offset as follows

$$\begin{aligned}\overline{H}_0(f,t)H(f,t) = (H_{s0} + \sum_{n_0 \in G_{m0}} \alpha_{n_0}e^{-j2\pi\frac{d_{n_0}(t)}{\lambda}})\times \\ (H_s + \sum_{n \in G_m} \alpha_n e^{-j2\pi\frac{d_n(t)}{\lambda}}).\end{aligned} \tag{5}$$

To remove phase offset, another common method is CSI ratio introduced in [4], [6]. The basic idea is to calculate the CSI ratio of two antennas. Since these methods have been well studied in previous work, we do not introduce CSI ratio in detail. The conjugate multiplication results with respect to different behaviors are shown in Fig. 2, where the mapping relationship between the moving of the human body and signal changes can be clearly observed.

B. Threat Model

The adversary's aims to collect indoor Wi-Fi signals to spy on indoor user activity without being perceived. In this paper, we define the threat model as follows.

Adversary Devices. The Wi-Fi device of the adversary is equipped with multiple antennas. By calculating the CSI ratio or CSI conjugate multiplication, the adversary can eliminate time-variant random phase offset for signal analysis.

Through-wall Wireless Sensing. The adversary can not enter the room, and thus cannot directly grasp the situation inside. The adversary collects the signal that travels through the wall rather than a direct LoS signal. The room is built with regular building materials, instead of expensive materials capable of absorbing Wi-Fi signals.

Eavesdropping-based Sensing. The adversary infers the activity of the user via eavesdropped signals. That is, the adversary only has the receivers and does not actively send signals to realize through-the-wall sensing. As illustrated in Fig. 4(a), when both the transmitter and receiver are outdoors, the sensing range is limited, as discussed in [19]. In contrast, when the adversary eavesdrops on the indoor Wi-Fi signal, the received signal is more sensitive to indoor activity, and the sensing range is shown in Fig. 4(b) [19]. Hence, the adversary's sensing area is expanded by the indoor transmitter.

III. Basic Model

A. Adversary View

The purpose of the adversary is to infer user actions or locations from the CSI readings. As such, the basic idea is that the amplitude and phase difference of CSI will change due to human body movement, and there is a potential mapping relationship between these changes and the user's behavior. Pattern-based and model-based methods can be employed in realistic environments to explore this mapping relationship.

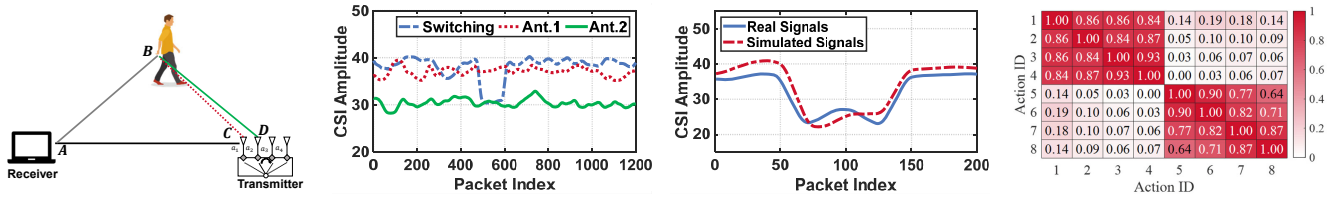
The pattern-based approach is currently a popular solution in wireless sensing. These methods use machine learning methods to find the mapping relationship between CSI readings and human behaviors [20], [21]. However, pattern-based approaches face the following two disadvantages: First, pattern-based approaches rely on collecting user data for pre-training and stable signal patterns for recognition, which undoubtedly requires more efforts. Second, wireless signals are very sensitive to environmental changes. Even for the same human behavior, there might be different signal patterns from location to location. Hence, pattern recognition using raw CSI readings is not suitable for cross-domain scenarios [22].

The model-based approach generally realizes wireless sensing by revealing the mathematical relationship between the CSI readings and the behaviors. The insight of the model-based approach is as follows: As illustrated in Fig. 3, the signal path length reflected off the human body will change when the users move or change their postures. According to the Doppler effect, the change in reflected path length will affect the frequency of the reflected signal, and we have

$$f_{Doppler} = f\frac{v_{path}}{c}, \tag{6}$$

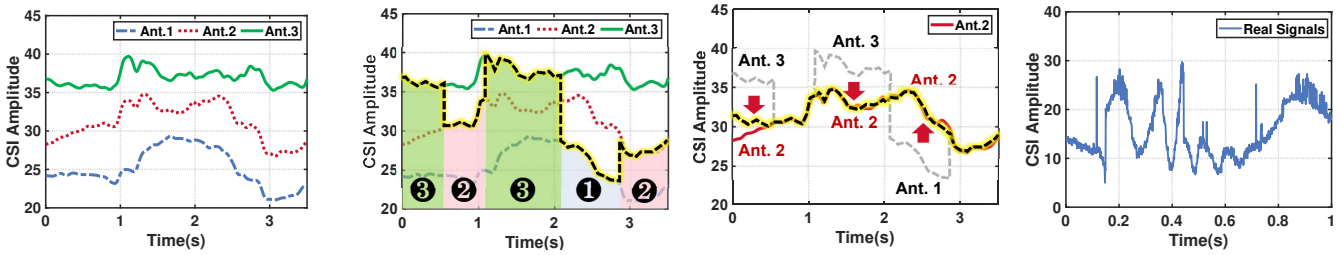
where $f_{Doppler}$ is the Doppler frequency shift (DFS), f is the communication frequency of the signal, v_{path} is the path length change rate (PLCR), c is the propagation speed of the Wi-Fi signals. Consequently, if we can derive the DFS from the raw CSI readings based on short-time Fourier transform (STFT), it is feasible to infer how the reflected path length changes. It is the basic idea of the model-based approach.

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(a) (b) (c) (d)

Fig. 5: (a) The basic idea of antenna switching; (b) The antennas switching can change the received signals; (c) We generate a "fraud" signal to block adversary; (d) The similarity of "fake" signal and real signal.



(a) (b) (c) (d)

Fig. 6: (a) Different antennas share the same signal pattern; (b) Switching antennas can change the pattern of the signal; (c) The process of signal recovery; (d) Raw data contains a mix of different kinds of mutations.

Accordingly, our protection system should be feasible for both pattern-based and model-based methods.

B. Protection Goal

Secur-Fi acts as a multi-tasker. First, Secur-Fi is able to generate a jamming signal to prevent adversaries from monitoring the situation of indoor users; Second, indoor authenticated users can still perform accurate wireless sensing under the above conditions. Next, we will introduce the basic insight to achieve such a goal.

First, our insight to prevent outdoor adversaries from monitoring is illustrated in Fig. 5(a). In particular, we connect the transmission link and multiple antennas with a switcher. When we switch different antennas to emit the signals, it is able to observe different signal features at the receiver. We conduct experiments to verify this assumption, and the experimental results are shown in Fig. 5(b). Under the static situation, the CSI readings at two antennas are stable. However, when we switch the transmission antennas, we are able to observe a changing signal even at static scenarios.

This motivates us to think about whether we could generate a "fraud" action signal by designing the switching rules, *i.e.*, behavior codebook in this paper, so that such signal can be considered to come from real behavior. As shown in Fig. 5(c), the red line is the signal with respect to the real user action, while the blue line is our fraud signal generated by switching antennas. Two signals are so similar that we can hardly distinguish them. More experimental results are shown in Fig. 5(d), where we list the correlation coefficient in the table. In particular, the correlation coefficient indicates how similar different signals are. Among them, $1^{st} \sim 4^{th}$ signals are corresponding to the bending action, while $5^{th} \sim 8^{th}$ signals are corresponding to the squatting action. Besides, $3^{rd}, 4^{th}, 7^{th}, 8^{th}$ are our generated fraud signals via antennas switching, while $1^{st}, 2^{nd}, 5^{th}, 6^{th}$ are real signals caused by real human behavior. The experimental results show a high similarity between fraud and real signals. Hence, it is feasible to prevent the adversaries from monitoring via the proposed system. However, since such changing signals can also interrupt the normal wireless sensing of the authenticated users, it is necessary to design a mechanism to extract correct signal features for indoor users.

Second, in order to enable authenticated wireless sensing, our insights are as follows. As shown in Fig. 5(a), when the user performs daily activities, the direct path of a_1 can be denoted as $|AB|$, the reflected path is $|AB|+|BC|$, while the direct path of a_2 can be denoted as $|AD|$, the reflected path is $|AB| + |BD|$. Since the distance $|CD|$ between the two antennas is much smaller than the length of the propagation path, the length change of the reflected paths with respect to the two antennas is similar. Consequently, we can observe similar signal patterns across different antennas, as shown in Fig. 6(a). If we perform the antenna switching during the user's action, we can observe significant changes in the features of the action signal compared to the original signal. As shown in Fig. 6(b), the black line is the ideal jamming action signals via antenna switching, which do not correspond to real action. Nevertheless, as illustrated in Fig. 6(c), if the mutations are removed according to the behavior codebook, the recovery signal (labeled with the black line) is similar to the original signal (labeled with the red line). The correlation coefficient represents the similarity between the recovered signal and the signals across the three antennas, and the average correlation

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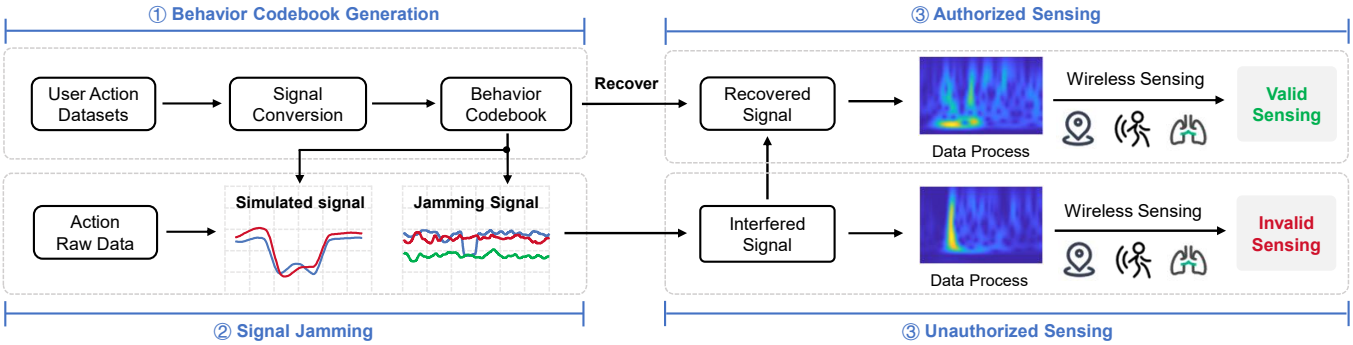


Fig. 7: System overview.

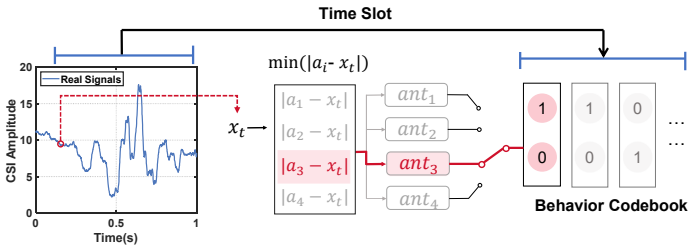


Fig. 8: Behavior codebook generation.

coefficient is 0.88.

A natural question is whether the adversary can detect the mutation so that they can recover the signal without the codebook. The answer is negative because there are many mutations in the original signal even without switching antennas in Fig. 6(d). An adversary cannot distinguish whether the change is caused by an action, switching antennas, or even environmental noise. Hence, the adversary cannot eliminate the effects caused by the antenna switching.

To summarize, there are two distinct observations: 1) For unauthorized outdoor adversaries, our system designs two approaches to interfere with the signal. First, we can generate a fraud signal by switching antennas to enable the adversaries to observe fraud user behavior, even when no one is at home; Second, we can switch antennas during the user's action to realize signal jamming, which prevents attackers from extracting correct actions. 2) For indoor authorized receivers, we utilize the behaviour codebook to eliminate the effect of antenna switching on the CSI, and then perform authorized wireless sensing.

IV. System Design

A. System Overview

The workflow of Secur-Fi is illustrated in Fig. 7, which consists of the following stages:

Behavior Codebook Generation. The behavior codebook refers to the rules for switching antennas to generate fraud motion signals or destroy the regular signals. The basic principle is that CSI readings with respect to multiple antennas are different. We will introduce how to transform the signals into the control logic of the switcher.

Signal Jamming. We perform antenna switching to generate action signals when the user does not perform action; when the user performs some definite actions, antenna switching not only interfere with the action signal but also regular sensing. Next, we will discuss how to recover the signal for indoor authorized sensing.

Signal Recovery. Since the behaviour codebook is available for the indoor authorized user, we design a method to eliminate the interference caused by the antenna switching. Based on this, we can realize regular wireless sensing. In contrast, without the behavior codebook, an outdoor adversary cannot infer the user's action.

B. Design Details

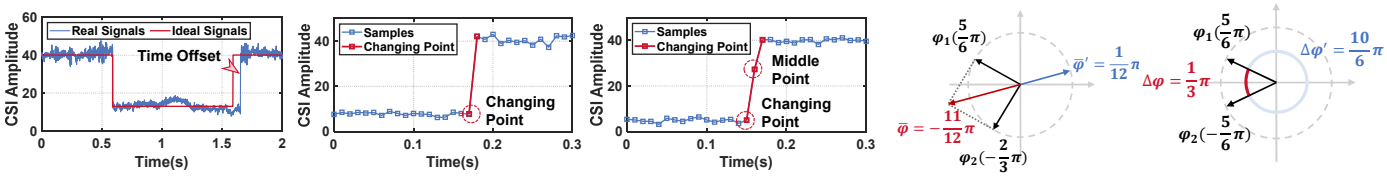
1) Behavior Codebook Generation: The behavior codebook is a combination of codes that control the active state of the antennas. It has two modes, which consist of action simulation and signal jamming.

Action Simulation. The purpose of action simulation is to generate a fraud signal to deceive outside adversaries. Our basic idea comes from *Quantization*, where we replace the CSI amplitude at a given moment with the closest antenna amplitude. As shown in Fig. 8, we measure the amplitude of four antennas in advance, and define them as $\boldsymbol{a} = [a_1, a_2, a_3, a_4]$. If the amplitude of an given action sequence is $\boldsymbol{x} = [x_1, x_2, ..., x_n]$, the operating state of four antennas can be represented as a series of 2-bit **codes**. Taking time slot x_t as an example, we compare $\boldsymbol{x_t}$ with $\boldsymbol{a}$, and determine which $\boldsymbol{a_i}$ is closest to $\boldsymbol{x_t}$. Next, we convert such $\boldsymbol{a_i}$ into 2-bit codes such as $[1, 0]$. By processing the whole action sequence in the same way, we can obtain the code combinations corresponding to the action sequence as the guide for antenna switching.

Signal Jamming. We can switch the antenna randomly when the user performs actions. Since the antenna switching does have a considerable impact on the signal, such random switching will interfere with the wireless sensing of outdoor adversaries. For this purpose, we can design a trigger signal. In particular, when the user performs an action, we will observe a significant signal energy. At the same time, we set the trigger signal as "true" and switch the antenna randomly. In this way, we can realize signal jamming.

2) Behavior Codebook Based Filter Method: The methods mentioned above can also affect the received signal of authorized users. Hence, we propose a behavior codebook-based filtering method to eliminate the effect of deliberate jamming

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(a) (b) (c) (d) (e)

Fig. 9: (a) There is a slight bias between the preset switching time point and the actual switching time point; (b) Instantaneous switching; (c) Non-instantaneous switching; (d) The average of phase; (e) The distance of phase.

on the sensing. However, it is non-trivial to realize such goal due to the following problems:

Problem 1: Phase Jump. The changes in amplitude and phase caused by the switching between antennas are the core basis for determining the mutation point. However, the phase jump can hide the preset changes and does not help identify the actual antenna mutation. When the phase is close to $-\pi$ or π, the phase jump occurs repeatedly between $-\pi$ and π, as shown in Fig. 10(a). The phase jumps need to be removed in order to recover the signal accurately.

Problem 2: Clock Bias. Though the behavior codebook specifies the working schedule of the antennas, we find that the hardware devices cannot work precisely the same time as the behavior codebook defines. As shown in Fig. 9(a), there is a slight time bias between the actual switching time and our schedule, and the time bias is usually within $\pm 0.08s$ (about 80 packets). Since daily action often lasts $1 \sim 3$ seconds [23], such short time bias does not drastically affect the simulation of action signals. Nevertheless, if we still follow the behavior codebook strictly for signal recovery, there will undoubtedly be a considerable difference between our recovered signals and the actual signals without interference.

Problem 3: Non-instantaneous Switching. As shown in Fig. 9(b), ideally, when we switch the antenna, the CSI readings should change within one packet, which we called as instantaneous switching. Nevertheless, the actual signal does not change immediately when we switch the antenna, as illustrated in Fig. 9(c). This non-instantaneous switch requires us to deal with a period of data instead of one point in the case of recovering the signals, which is challenging to realize.

To address the above problem, our solutions are as follows:

Solution 1: Circular Standard Deviation. In order to recognize the phase jump, we propose the circular standard deviation (CSD). In contrast to standard deviation, the advantage of our proposed CSD is to deal with circular jumps of phase information. To this end, we begin with the calculation method of the standard deviation as follows:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(\varphi_i - \overline{\varphi})^2}{N}}, \tag{7}$$

which can be divided into two parts: how to calculate the average value $\overline{\varphi}$, and how to calculate the difference $\varphi_i - \overline{\varphi}$.

First, to determine $\overline{\varphi}$, the common arithmetic methods of calculating the average are not feasible, because the algebraic averaging is affected by periodicity. For example, for two angles $(-\frac{2}{3}\pi, \frac{5}{6}\pi)$, the algebraic sum is $(-\frac{2}{3}\pi + \frac{5}{6}\pi)/2 = \frac{1}{12}\pi$ in Fig. 9(d). Obviously, it is unreasonable to obtain such a result. To address this issue, we might as well represent them in the polar coordinate system, and calculate the vector sum. In this way, we can obtain the desired vector $\overline{\varphi} = -\frac{11}{12}\pi$ for two angles $-\frac{2}{3}\pi$ and $\frac{5}{6}\pi$. Therefore, for the whole sequence, we have

$$\overline{\varphi} = angle(\sum_{n=1}^{N} e^{j\varphi}), \tag{8}$$

where $angle(\cdot)$ indicates the operation to obtain the phase of complex exponential number while $angle(e^{j(2k\pi+\phi)}) = \phi, \{k \in Z, \phi \in [-\pi, \pi]\}$.

As for the $\varphi_i - \overline{\varphi}$, we need to choose the appropriate angle of intersection of the two vectors as illustrated in Fig. 9(e). Take $(-\frac{5}{6}\pi, \frac{5}{6}\pi)$ as examples, there are two results $\Delta\varphi = \frac{1}{3}\pi$ and $\Delta\varphi' = \frac{10}{6}\pi$. It is obvious that the phase we desire is between $-\pi$ and π. The function $angle(\cdot)$ can bring us the appropriate angle. Consequently, it can be expressed as

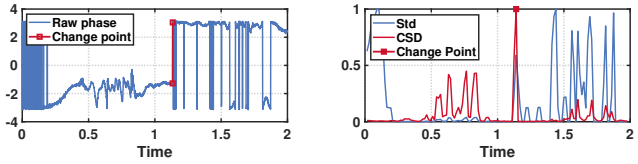
$$\Delta\varphi_i = angle(e^{j(\varphi_i - \overline{\varphi})}). \tag{9}$$

Based on the Eq. 7, Eq. 8 and Eq. 9, we can obtain the circular standard deviation as follows

$$\sigma_{CSD} = \sqrt{\frac{\sum_{i=1}^{n}(min(|\varphi_i - \overline{\varphi}|, 2k\pi - |\varphi_i - \overline{\varphi}|)^2}{N}}. \tag{10}$$

In this way, we can remove the disturbance of phase jump on the mutation detection. As illustrated in Fig. 10(b), we can identify the change point correctly based on the CSD. In contrast, the normal STD can not identify such change point.

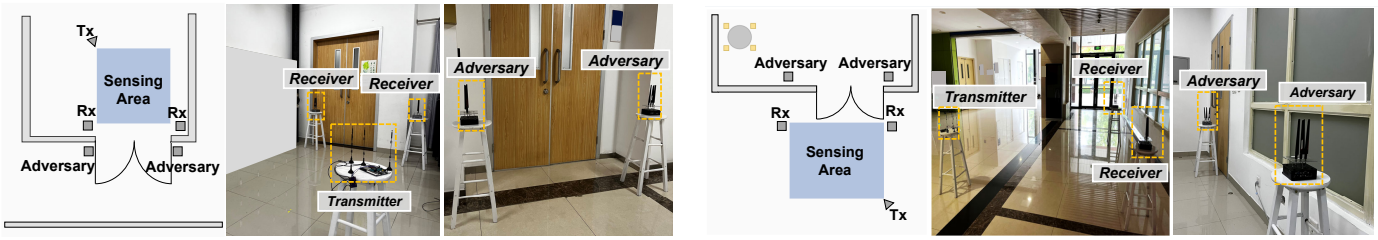
Solution 2: Dual Sliding Window Based Filter. The hardware bias causes the behavior codebook to be used only as a time range for antenna switching, not as the exact points in time as illustrated in Fig. 9(b) and (c). To realize range-based switching recognition, we propose a dual sliding window



(a) (b)

Fig. 10: (a) The phase jumps hide the phase change caused by antenna switching. (b) Circular standard deviation can identify the antenna switching time.

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(a) Office (b) Open Space

Fig. 11: Experimental Environment

Algorithm 1 Sliding Window Based Filtering Algorithm

Input: Raw Channel state information C, behavior codebook $B = [t_1, t_2, ..., t_K]$, $t_{offset} = 200$

Output: Recovered Channel state information C_{Rec} and phase conjugate multiplication P_{Rec}

Part A: Remove Time Offset

t_{offset};
for $k := 1; k < K; k++$ **do**
  for $i = t_k - t_{offset}; i < t_k + t_{offset}; i++$ **do**
    Construct the silding windows with Win_size and Step_size
    Find the index of the window with the maximum variance denoted as **W[k]**
    for $j = W[k]; j < W[k] + Win_size; j++$ **do**
      Construct silding windows with win_size and step_size
      Find the window with the largest variance denoted as **T[k]**
    end for
  end for
end for

Part B: Recover Change

for $k = 1; k < K; k = k + 2$ **do**
  for $m = T[k]; m < T[k+1]; m++$ **do**
    Calculate the gap before and after T[k].
    Recover the signal from T[k] to T[k+1].
  end for
end for

based filtering method. Intuitively, the first sliding window is used to detect the degree of data fluctuation, while the second sliding window is used to recover non-instantaneous switching. For this purpose, we set the first sliding window as $0.02s$ (20 packets for 1000Hz packet rate) and the second sliding window as $0.002s$ (2 packets for 1000Hz packet rate). 1) In the first sliding window, we select the adjacent 200 sample points according to the switching time of the codebook for analysis. For these readings, we calculate the variance within $0.02s$ to determine the approximate location of the fluctuation. 2) In the second sliding window, our observation is that if an instantaneous switch occurs, there will be only one window exhibiting a larger variance compared with the others. Otherwise, more than one window will exhibit a more considerable variance compared with the others. Based on the above observations, we can determine whether the non-instantaneous switch occurs and the corresponding time. Finally, we compensate for the amplitude and phase changes caused by the antenna switching and thus obtain the recovered signal. The design details are illustrated in Algorithm 1.



(a) (b) (c)

Fig. 12: Experimental Devices

C. Application Scenarios and Methodology

According to previous work [4], [12], [17], current wireless sensing technology consists of main tasks including activity recognition, movement tracking and vital sign monitoring. We will respectively verify the performance of Secur-Fi based on the common activity recognition method for these categories. In the case of activity recognition application, we utilize the learning-based methods, including Dynamic Time Warping (DTW), Support Vector Machine (SVM), k-Nearest Neighbors (kNN), Decision Tree (DT) and Random Forest (RF). In the case of passive tracking applications, we utilize the model-based method established in [12]. In the case of respiration monitoring applications, we utilize the periodicity-based methods [4] to detect periodic signals.

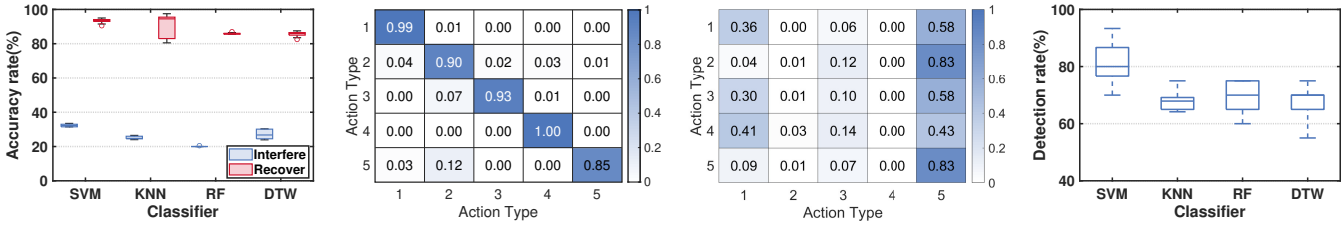
V. Implementation and Evaluation

A. Implementation and Metrics

Implementation. As shown in Fig. 12(a) and Fig. 12(b), we implement the proposed system on the HC6800-ES V2.0 micro-controller [24] and a switch board to transmit Wi-Fi signals. The micro-controller schedules the antenna switching according to the behavior codebook, and four antennas correspond to the 2-bit code in the behavior codebook. We utilize the csitool [25] to collect CSI of the off-the-shelf devices equipped with Intel 5300 NIC [25] in Fig. 12(c).

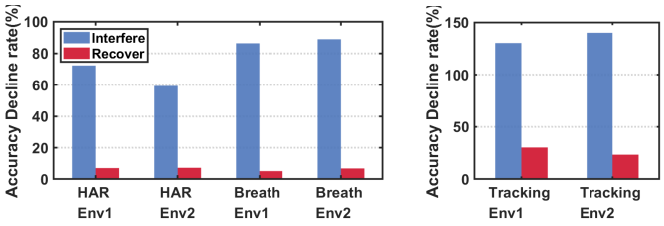
Experimental setup. The experiments are conducted in two different environments, including an office room and an open space, as illustrated in Fig. 11. During the experiments, there is always a wall between the user and the adversary. We conducted experiments with 7 volunteers, including 5 males and 2 females. The volunteers are required to perform daily activities, walk and breath within the sensing area.

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(a) (b) (c) (d)

Fig. 13: Comparison of the accuracy of different actions under various classifiers.



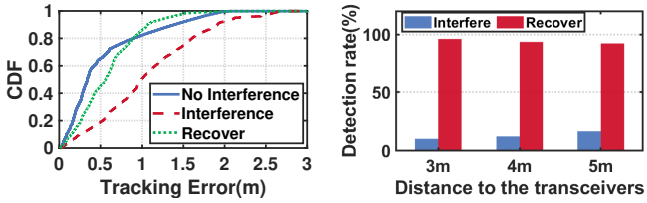
(a) Human action recognition and breath (b) Office

Fig. 14: Overall performance of Secur-Fi.

Metric. We define accuracy decline rate $\boldsymbol{\vartheta}$ as the metric to numerically measure the effectiveness of interference and recovery. It can be calculated as follows: $\boldsymbol{\vartheta} = (Acc_{ori} - Acc_{aft})/Acc_{ori} \times 100\%$, where Acc_{ori} is the accuracy without interference, Acc_{aft} is the actual accuracy with interference or recovery. For tracking applications, as we prefer a smaller tracking error, the above metric can be rewritten as $\boldsymbol{\vartheta} = (Err_{aft} - Err_{ori})/Err_{ori} \times 100\%$, where Err_{ori} is the accuracy without interference, Err_{aft} is the actual tracking error with interference or recovery. $\boldsymbol{\vartheta}_{out}$ depicts the degree of our interference method in the accuracy of the adversaries, while $\boldsymbol{\vartheta}_{in}$ depicts the degree of our recovery method in the accuracy of the authorized user.

B. *Overall Performance*

We first calculate $\boldsymbol{\vartheta}_{out}$s and $\boldsymbol{\vartheta}_{in}$s for all volunteers and average them as the final results, as shown in Fig. 14. It can be observed that the accuracy of activity recognition, passive tracking and respiration monitoring drops significantly in the interfered mode, which $\boldsymbol{\vartheta}_{out}$s are 72%, 130%, 86%, respectively. In contrast, after we recover the signal for the authorized user, the accuracy only drops by 7.2%, 23.3%, 5.1% for three applications, respectively. In order to avoid over-fitting and make full use of the raw data, we adopt the 10-fold cross-validations to randomly divide all the data into 10 identical subsets and select one of the subsets as the tests set each time and the others as the training sets in the case of activity recognition. The results demonstrate that Secur-Fi is effective in preventing outdoor eavesdropping while ensuring authorized sensing. Moreover, there is no significant difference between the two experiment environments, and the accuracy difference is only 2.4% on average. These results indicate that Secur-Fi is robust in blocking eavesdropping as well as ensuring authorized sensing.



(a) (b)

Fig. 15: Tracking and breath detection performance.

C. *Performance of Activity Recognition*

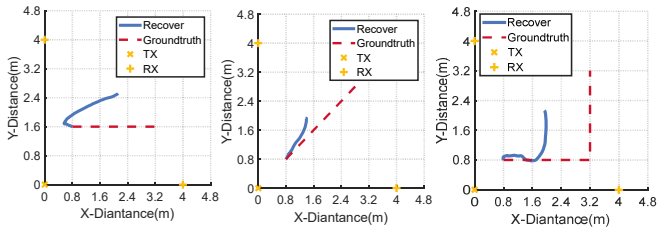
We design five gestures, including squatting, sitting, waving, jumping up and stepping forward. For each volunteer, the gestures are repeated 20 times in the state of non-interference and interference. Thus, there are $2 \times 7 \times 5 \times 20 = 1400$ experiments in all. For a fair comparison, ten features are extracted from both the time and frequency domains for the recovered data received from indoor and the data received from outdoor devices, including mean, variance, first quartile, third quartile, skewness, kurtosis, maximum, minimum, information entropy and maximum frequency domain [6], [7], [17]. Ultimately, we obtain the final evaluation performance by averaging the 10 cross-validations.

Fig. 13(a) shows the accuracy of action recognition for the interference and recovery modes. It is clear that the recognition accuracy of the interference mode decreases significantly under all classifiers, reaching a minimum of 31.5%, 24%, 20%, 23.9% under the SVM, KNN, RF and DTW, while the recognition accuracy can achieve a maximum of 94%, 96.5%, 86.5% and 87.5% in the recovery mode, respectively. Fig. 13(b) and (c) show the specific recognition confusion matrices of SVM, where the ID $1-5$ correspond to squatting, stepping forward, jumping, waving and sitting, respectively. The confusion matrices exhibit that the interference mode of Secur-Fi is capable of "converting" a certain action into other actions by switching antennas, while the recovery mode can maintain the normal human activity recognition. In the case of action simulation, we set the generated signal samples to the test set and average the results. As shown in Fig. 13, the simulation action detection rate of Secur-Fi can achieve 72.7% in average under five classifiers. This means that Secur-Fi is able to deceive an adversary into recognizing a fraud action even in a stationary environment.

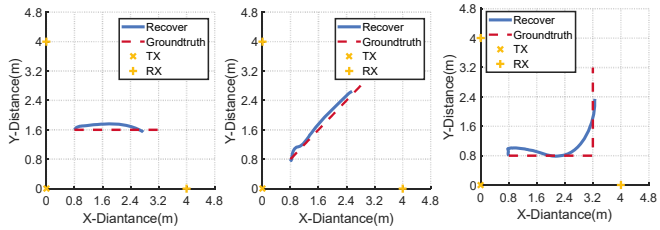
D. *Performance of Passive Tracking*

We utilize the model-based method to track device-free users and compare the performance after interference and

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(a) The traces after interfering



(b) The traces after recovering

Fig. 16: Tracking performance.

recovery. We use 2 receivers and the source codes provided by Widar [12] and Widar 2.0 [13] to realize passive tracking. For each kind of trace, volunteers are asked to walk for 8 times. In Fig. 16(a), it can be observed that switching antennas interfere significantly with tracking performance in terms of both speed and direction. As shown in Fig. 16(b), the experimental results show that ϑ_{out} is 130.2%. Moreover, the median tracking error with interference is $0.99m$, while the median tracking error without interference is $0.43m$. As for recovered pattern, ϑ_{in} is 16.2%, and the median tracking error is $0.56m$. Hence, it is hard for an adversary to obtain the user's trajectories.

E. *Performance of Respiration Monitoring*

We utilize the methods illustrated in FarSense [4] to detect the respiration cycle. We deploy a pair of transceivers with the LoS of $3.2m$, and ask the volunteers to sit in the chair located on the vertical bisector of the LoS path.

In Fig. 15(b), the detection rates of the interference mode of *Secure-Fi* from different distances to the LoS are significantly low. In contrast, detection rates under the recovery mode can still achieve over 96%. This illustrates that even slight behaviors such as breathing can be blocked by Secur-Fi from being eavesdropped on by adversaries.

VI. Related Work

When we benefit from novel IoT technologies, we should also focus on preventing new attacks. Based on this perspective, we will analyze the opportunities and threats of wireless sensing technology. There are three mainstream applications of wireless sensing, including passive tracking, behavior recognition and breath detection.

Passive Tracking. In such systems [10], [12], [13], [26]–[30], we usually extract position features from the raw CSI readings. These features including angle of arrival (AoA), time of flight (ToF) and Doppler frequency shift (DFS). They respectively depict the angle, distance and velocity of the user, based on which we can derive the trajectory of the user. The advantage of passive tracking is that we can track users who do not carry the device. However, as the technology of through-the-wall tracking [10], [31], [32] is gradually being broken through, the adversary from the outside might track us. Hence, how to protect the privacy of users has become a key problem.

Behavior Recognition. Device-free behavior recognition plays a vital role in human-computer interaction. For example, we are able to control the smart device through simple gestures easily [6], [15], [33], [34]. Moreover, fall recognition [35] helps us provide quick rescue for victims. Although these methods enable behavior recognition to be generalized to a variety of scenarios, it also increases the risk of being eavesdropped on by attackers. However, the existing research mainly focuses on improving the recognition precision while neglecting the security aspects. In WiCAM [36], the authors define the wireless attack imperceptibility for both communication and sensing and propose a digital adversarial perturbation to attack WiFi sensing system with minimized impact on WiFi communication. State of the art [37] finds that behavior recognition is vulnerable to manipulating physical signals, and they introduce two approaches to prevent authorized wireless sensing. However, they ignore that if our sensing signals have eavesdropped on, this could still lead to privacy breaches. This paper aims to introduce how to prevent outside eavesdropping while maintaining authorized wireless sensing.

Breath Cycle Detection. One might wonder whether we cannot be eavesdropped on when we do not perform gestures or move. In fact, Wi-Fi can also remotely detect our breathing cycle [4], [5], [38]. Specifically, Fullbreathe [38] reveals that there are certain locations called "blind spots", where the breath detection might experience poor performance. Accordingly, they design the method to eliminate these blind spots. Furthermore, FarSense [4] has increased the distance of breath detection to 8 meters. The art [19] discusses the effects of device placement for WiFi sensing, and their experiments also verify the feasibility of through-wall breath detection. To summarize, even if we are just breathing at home, eavesdroppers might use this technology to tell if someone is in the house. Consequently, it is necessary to develop a method to prevent unauthorized sensing of the adversary.

VII. Conclusions

In this paper, we proposed Secur-Fi, which can prevent outside eavesdropping while maintaining inside authorized sensing. We generate behavior codebooks to emit changing signals to block unauthorized sensing. According to the behavior codebook, authorized devices can distinguish whether actual user actions or antenna switches caused CSI changes. Experimental results in the real environments demonstrated that Secur-Fi can effectively reduce the accuracy of unauthorized sensing by 130% (passive tracking), 72% (behavior recognition), 86% (breath detection).

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